Figure 1. From transition scenarios to aggregate productivity: framework and validation

A Framework

Transition scenario outputs

Sectoral prices, quantities, costs, technology shares

Sector-level productivity mapping

Infer sectoral TFP from price and quantity changes

$\mathrm{TFP}_{i,h} = f(p_{i,h}, q_{i,h}; \zeta)$

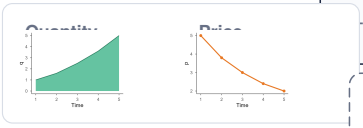
Domar aggregation

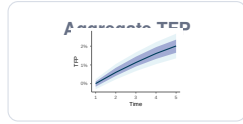
Weight sectoral TFP by Domar weights

$\mathrm{TFP}_h^{\mathrm{agg}} = \sum_i \lambda_i \, \mathrm{TFP}_{i,h}$

Aggregate TFP effect

Economy-wide productivity impact across scenarios



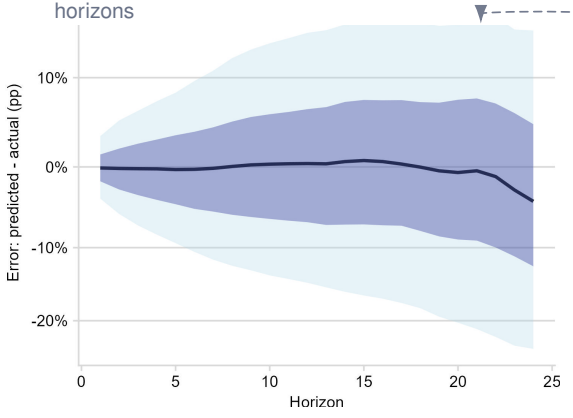


Historical calibration and validation

Calibrates ζ, evaluates forecast bias, quantifies uncertainty

B Industry-level hindcasting error

Predicted minus observed TFP growth across forecast horizons



C Aggregate hindcasting error

Error after Domar aggregation across forecast horizons

